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Summary of Learning

In this assignment I learned how end-to-end delay in a network is composed of transmission delay, propagation delay and queuing delay. I also learned how store-and-forward routers work and how packets accumulate in queues when the arrival rate is faster than the service rate. By computing the delay packet-by-packet I saw that the slowest link determines congestion and causes queue growth. Using Docker and tc commands I was able to simulate bandwidth limits and propagation delays and verify the theoretical results with packet captures.

Challenges Faced

The main difficulty was that I initially misread the assignment document, which caused me to complete the calculations incorrectly and redo the assignment. After reviewing the assignment requirements again, I realized the mistake and corrected my work. After that, I reviewed the store-and-forward behavior and fixed the table so that only packets waiting for transmission were considered queued. Another challenge was setting up Docker and clearing old tc configurations, which caused incorrect results until the environment was reset.

Time Spent

Approximately 6-8 hours total including setup, calculations, debugging and verification.

Packet Drop Explanation

In the computed table no packets were dropped because router queue sizes were a=9, b=8 and c=5, which were sufficient to store all packets before transmission. However, during the live network experiment some packets may still drop due to timing differences, scheduling overhead and implementation behavior in the operating system network stack.

Network Delay Explanation

Packets were transmitted every 25 ms from Host A but routers forwarded at slower rates (50 ms, 100 ms, and 125 ms). Because the arrival rate exceeded the service rate, packets accumulated in router buffers causing increasing queuing delay for early packets. After the initial burst, packets were spaced farther apart and queues drained, leading to stable delays. The last bit of the final packet arrived at Host B at 1650 ms.

Commands Used

Host A:
tc qdisc add dev eth0 root handle 1: htb default 10
tc class add dev eth0 parent 1: classid 1:10 htb rate 400kbit ceil 400kbit
tc qdisc add dev eth0 parent 1:10 handle 10: netem delay 10ms

Router R1:
tc qdisc add dev eth1 root handle 1: htb default 10
tc class add dev eth1 parent 1: classid 1:10 htb rate 200kbit ceil 200kbit
tc qdisc add dev eth1 parent 1:10 handle 10: netem delay 40ms limit 9

Router R2:
tc qdisc add dev eth1 root handle 1: htb default 10
tc class add dev eth1 parent 1: classid 1:10 htb rate 100kbit ceil 100kbit
tc qdisc add dev eth1 parent 1:10 handle 10: netem delay 20ms limit 8

Router R3:
tc qdisc add dev eth1 root handle 1: htb default 10
tc class add dev eth1 parent 1: classid 1:10 htb rate 80kbit ceil 80kbit
tc qdisc add dev eth1 parent 1:10 handle 10: netem delay 30ms limit 5